SAFEDRIVE INSURANCE LTD.

INSURANCE CLAIMS ANALYTICS AND PREDICTIVE RISK ASSESSMENT

End-to-end Capstone Business Report

Portfolio policies	Observed claims	Overall claim rate	Mean NCAP rating
58,592	3,748	6.40%	1.76 / 5

Purpose

Analyse historical motor insurance policy, policyholder, vehicle and safety-feature data; understand observed claim patterns; compare predictive models; and assess whether a risk score could help prioritise policies for human review.

Decision statement

The selected model improves ranking performance relative to the Logistic Regression baseline, but low precision means it is not suitable for automated underwriting or claim decisions. Any use should remain a controlled, human-reviewed pilot.

Prepared from the three supplied capstone notebooks and the ten dashboard screenshot assets supplied for this report. Reported figures reflect saved notebook outputs; refresh and reconcile Power BI exports before final submission.

Contents

Section	Coverage
Executive summary	Problem, headline findings, model result and decision recommendation
1. Business understanding	Business problem, objectives, questions and success metrics
2. Data and preparation	Dataset scope, target imbalance, quality checks and preprocessing
3. Exploratory analysis	Portfolio distribution, tenure, vehicle age, NCAP, segment and speed-alert availability
4. Feature engineering	Derived features, correlation screening and Featuretools interactions
5. Model development	Baseline Logistic Regression, validation and tuned Decision Tree comparison
6. Risk segmentation	Held-out predictions, risk bands, observed claim rates and business interpretation
7. Explainability and governance	SHAP, MLflow, model versioning and Evidently
8. Business recommendations	Suggested controlled workflow, monitoring and validation
9. Limitations and final submission checks	Risks, interpretation cautions and reproducibility checks
Appendix A	All supplied dashboard visual images with short captions

Executive summary

SafeDrive Insurance Ltd. seeks to understand claim patterns in its motor insurance portfolio and assess whether historical policy and vehicle attributes can help identify policies that may warrant closer review. The supplied dataset contains **58,592 policies**, of which **3,748 had a claim**, giving an overall observed claim rate of **6.40%**. The target is highly imbalanced: approximately 93.6% of records are no-claim policies.

The project follows a three-stage workflow: (1) data preprocessing, exploratory analysis and feature engineering; (2) Logistic Regression baseline modelling and validation; and (3) Featuretools-assisted feature generation, Optuna tuning, model comparison, SHAP explainability, experiment tracking, model versioning, data-drift reporting and test-set risk segmentation.

Headline measure	Result	Interpretation
Portfolio size	58,592 policies	One row per unique policy ID in the supplied data checks.
Observed claims	3,748 (6.40%)	Positive class is a small minority; accuracy alone is insufficient.
Selected model	Featuretools-Extended Decision Tree	Selected on mean 5-fold CV ROC-AUC.
Cross-validation ROC-AUC	0.6365 vs 0.6101	Tree outperformed Logistic Regression on ranking ability.
Held-out test ROC-AUC	0.6391	Some discrimination, but not strong enough for autonomous decisions.
Held-out test recall	73.07%	Detected about 73% of claim cases at the chosen class threshold.
Held-out test precision	8.79%	Many flagged policies are false positives; human review is essential.
Risk-band gradient	3.46% Low to 14.36% Very High	Observed claim rates increase across the saved notebook risk bands.

Overall recommendation: treat the model as an analytical prioritisation prototype, not a production-ready underwriting engine. Before submission, ensure the final Power BI data exports are refreshed from the final Stage 3 notebook and that dashboard counts, category labels and risk-band metrics reconcile exactly.

Evidence convention: quantitative statements are taken from the supplied notebook outputs or supplied dashboard screenshots. Interpretations and recommendations are explicitly framed as analysis, not causal claims or verified business outcomes.

1. Business understanding

1.1 Business problem

The Stage 1 notebook frames the business problem as SafeDrive having observed rising insurance claim payouts and wanting to identify policyholders more likely to file a claim during the policy period. The dataset does not itself quantify the trend in payouts over time, so this report treats that statement as the business context supplied by the project rather than an independently measured trend.

1.2 Analytical objective

Analyse historical insurance policy, policyholder, vehicle and safety-feature information; identify patterns associated with the binary target **is_claim**; build and validate a baseline classifier; compare it with a tuned Decision Tree; and translate the model outputs into risk bands and management-facing visuals.

1.3 Business questions

- How does observed claim rate vary across policy tenure, vehicle age, policyholder age, vehicle segment, NCAP rating and safety-system availability?
- Does the Featuretools-extended Decision Tree improve claim-risk ranking relative to Logistic Regression?
- Do the model's held-out risk bands show different observed claim rates?
- Which features most influence predictions, and what governance controls are required before any operational use?

1.4 Success metrics

Metric family	Measure	Why it matters
Business	Claim capture rate in a prioritised group	Shows how many observed claims fall within the group selected for review.
Business	High-risk segment lift	Compares the selected group's observed claim rate with the 6.40% portfolio rate.
Business	Investigation efficiency	Balances the number of policies reviewed against claims identified.
Model	Recall	Share of actual claims identified.
Model	Precision	Share of model-flagged policies that actually had a claim.
Model	F1 score	Harmonic balance between precision and recall.
Model	ROC-AUC	Ranking/discrimination across classification thresholds.
Governance	Drift, explainability and versioning	Supports repeatability and monitoring; does not prove production readiness.

The project does not define financial cost values for false positives or false negatives. Therefore, no monetary return on investment or claim-cost savings is asserted.

2. Data scope, quality and preparation

2.1 Dataset profile

Item	Finding from supplied notebooks
Rows	58,592
Raw columns	44
Engineered Stage 1 columns	49
Data types in raw dataset	28 object, 12 integer, 4 float columns
Missingness	Stage 1 output reports no missing values in the listed columns.
Duplicate records	Notebook's final checks are intended to verify duplicates; unique policy IDs are reported.
Unique policy IDs	58,592
Target	is_claim: 0 = no claim, 1 = claim
Target counts	54,844 no claim; 3,748 claim
Claim rate	6.40%
Train/test split	80/20 stratified; train 46,873, test 11,719
Split claim rate	Approximately 6.4% in both sets

2.2 Data preparation and engineered variables

The Stage 1 workflow inspects schema and data types, checks missing values and target distribution, examines numerical and categorical features, parses engine specifications, and constructs additional features for downstream modelling. The final Stage 1 data contains 49 columns and is used as the input to the modelling stages.

Feature	Description and interpretation
max_power_bhp	Numeric engine power parsed from the max_power specification.
max_torque_nm	Numeric engine torque parsed from the max_torque specification.
safety_equipment_count	Count of selected binary safety/monitoring features.
power_to_weight	Engine power divided by gross vehicle weight; confirm units before operational use.
High_Risk_Driver_Flag	Project-defined exploratory flag; not a validated underwriting rule.
policy_id	Identifier retained for traceability and joins, excluded from predictive features.

2.3 Outlier and multicollinearity handling

Stage 1 investigates outliers using the interquartile range (IQR) method and explicitly states that the original values are retained in the Stage 1 dataset. Any modelling-time capping or scaling should be fitted using training data only to avoid test-set leakage. The report does not claim that all outliers were capped, because the documented Stage 1 approach is an investigation and treatment plan rather than blanket deletion or replacement.

The notebook also uses correlation screening and variance inflation factor (VIF) checks on a reduced numeric feature set. These checks support identifying potential redundancy; they do not establish that every feature is independent or that all possible multicollinearity has been eliminated.

Stage 2 uses a stratified 80/20 split with a fixed random state and puts imputation, scaling and categorical encoding inside a modelling pipeline. This helps keep preprocessing learned from training data rather than the held-out test set.

3. Exploratory data analysis and portfolio findings

3.1 Overall claim distribution

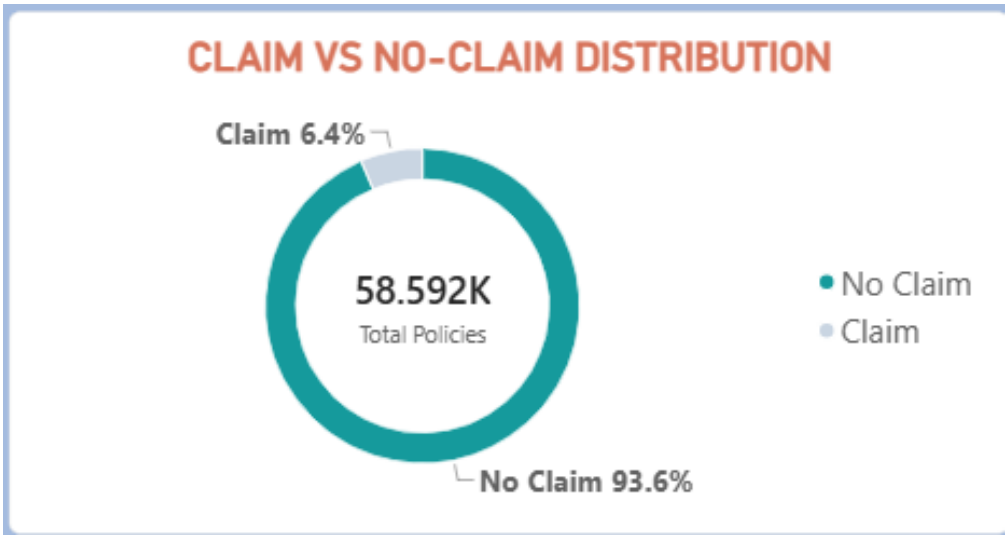


Figure 1. Claim vs no-claim distribution. The donut chart shows that 6.4% of portfolio records have a claim and 93.6% do not; the class imbalance motivates using recall, precision, F1 and ROC-AUC alongside accuracy.

The observed claim rate is 6.40%. For every claim record, there are approximately 14.6 no-claim records. A model can therefore appear accurate by favouring the majority class while still missing claims; model assessment must focus on the minority class and ranking performance.

3.2 Portfolio KPIs



Figure 2. Portfolio KPI cards. The visuals summarise 58,592 policies, 3,748 claims, a 6.40% overall claim rate, 11,719 scored test policies and average NCAP rating of 1.76.

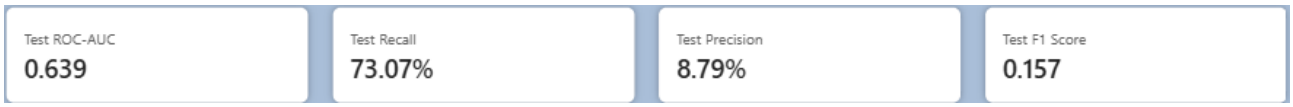


Figure 3. Held-out model KPI cards. The selected model reports ROC-AUC 0.639, recall 73.07%, precision 8.79% and F1 score 0.157 on the saved test evaluation.

3.3 Policy tenure

The Stage 1 notebook bins policy_tenure into five groups. The observed claim rate rises across the first four groups, then eases slightly in the highest-tenure group. This is a non-linear association and should not be interpreted as tenure causing claims.

Policy tenure group (ascending)	Policies	Observed claim rate
Q1 / shortest	11,719	3.58%
Q2	11,718	4.74%
Q3	11,718	6.66%
Q4	11,718	8.65%
Q5 / longest	11,719	8.35%

3.4 Vehicle age

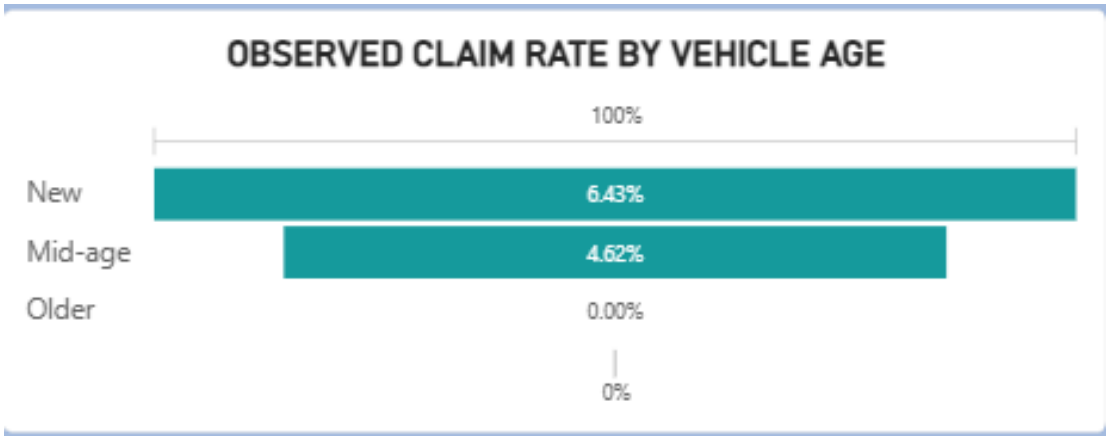


Figure 4. Observed claim rate by vehicle-age grouping. The displayed Older group has a 0.00% observed claim rate but only 14 policies; this is too small to support a reliable inference.

The Stage 1 quintile analysis reports vehicle-age group claim rates from 5.04% to 6.94%. The three-group Power BI visual is a separate grouping and shows an Older group with only 14 policies. The small-group rate should be interpreted cautiously; it is not evidence that older vehicles are inherently safer.

3.5 NCAP rating

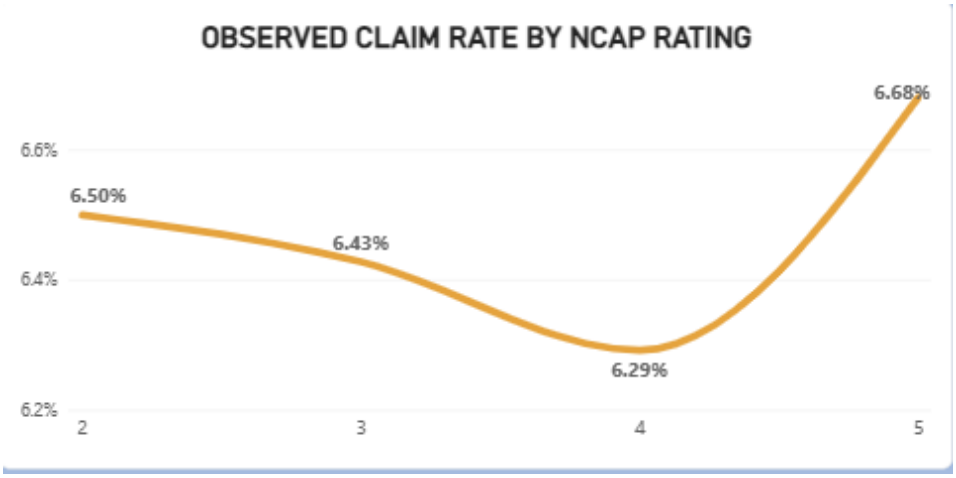


Figure 5. Observed claim rate by NCAP rating. The supplied visual shows rates around 6.29%–6.68% across ratings 2–5, with the highest displayed rate at rating 5.

The displayed rates vary modestly across NCAP ratings: 6.50% at rating 2, 6.43% at rating 3, 6.29% at rating 4 and 6.68% at rating 5. The line is a descriptive summary; it does not establish that a rating causes a higher or lower claim probability. Confirm the category counts before making decisions from small subgroups.

3.6 Vehicle segment

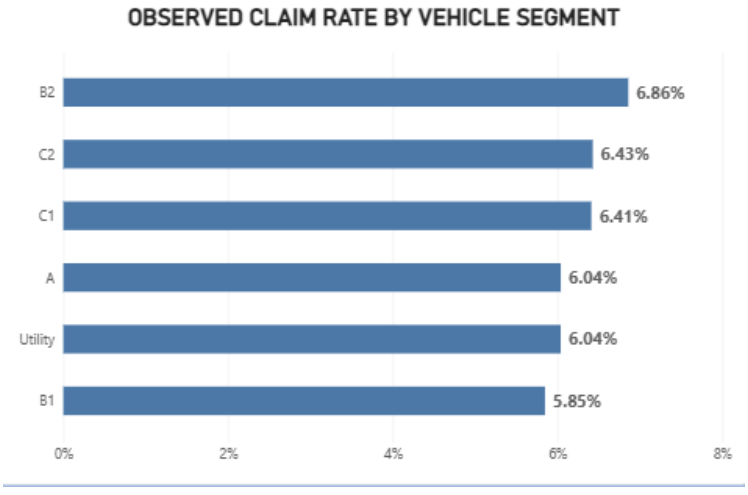


Figure 6. Observed claim rate by vehicle segment. Segment B2 has the highest displayed rate (6.86%) and B1 the lowest (5.85%); the differences are descriptive, not causal.

Vehicle segment	Observed claim rate in supplied visual
B2	6.86%
C2	6.43%
C1	6.41%
A	6.04%
Utility	6.04%
B1	5.85%

3.7 Speed-alert system availability

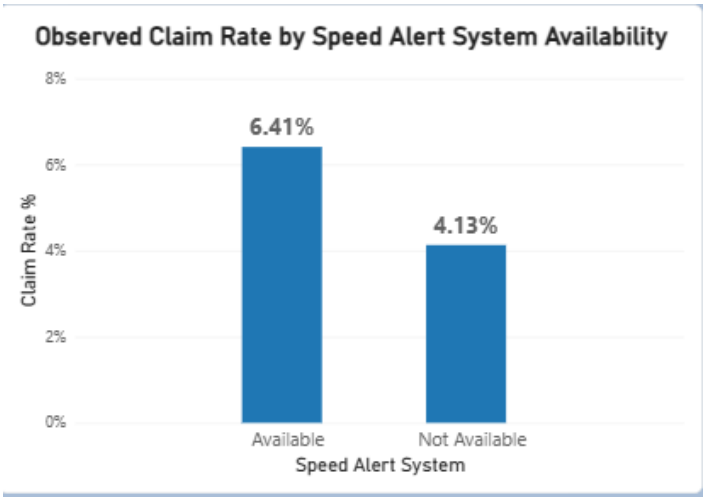


Figure 7. Observed claim rate by speed-alert system availability. Policies with the system available show 6.41%; policies without it show 4.13%. The field describes availability, not observed speeding incidents.

The speed-alert comparison should be interpreted carefully because the feature represents system availability rather than speeding behaviour, and the two groups are highly imbalanced in size. The chart alone does not justify concluding that the system increases or decreases claim risk.

4. Feature engineering and automated feature generation

4.1 Stage 1 feature engineering

The preprocessing notebook constructs business-oriented features to support exploratory analysis and modelling. These engineered columns supplement the original policy and vehicle attributes; they should be treated as candidate predictors whose value must be assessed through validation.

4.2 Featuretools Deep Feature Synthesis

Stage 3 reports 37 numeric base features and 1,332 generated Deep Feature Synthesis (DFS) candidates. The strongest base numeric target correlation was 0.08266. DFS candidates were screened against this reference, and two interactions were retained for the extended model:

Retained feature	Absolute correlation reported	Interpretation
cylinder × policy_tenure	0.083032	Small improvement over the strongest base numeric correlation.
policy_tenure × width	0.082688	Also slightly above the base numeric correlation reference.

The final extended feature set contains 47 features: 45 original predictors plus the two retained DFS interactions. Correlation screening is a simple filter and does not guarantee that selected interactions improve out-of-sample performance; this is evaluated in cross-validation and held-out testing.

4.3 Why the generated interaction matters

The interaction cylinder × policy_tenure appears as the strongest feature in the supplied SHAP summary. This is consistent with the model using a combined vehicle-engine and policy-tenure signal. It does not mean either variable causes claims, and it does not prove that the interaction is stable across future portfolios.

5. Model development and validation

5.1 Stage 2: Logistic Regression baseline

Stage 2 builds a Logistic Regression baseline using 45 predictors. The data is split into 46,873 training rows and 11,719 test rows, with stratification preserving the approximately 6.4% claim rate in each subset. The pipeline handles numerical imputation/scaling and categorical encoding.

Baseline test metric	Saved notebook output
Accuracy	0.5688 (56.88%)
Precision	0.0802 (8.02%)
Recall	0.5480 (54.80%)
F1 score	0.1399
ROC-AUC	0.5856

The baseline confusion-matrix interpretation reports 6,255 true negatives, 4,714 false positives, 339 false negatives and 411 true positives. This means it identified about 55% of actual claim cases, while flagging a large number of no-claim policies as potential claims. That false-positive burden would create a large review workload if used without threshold tuning.

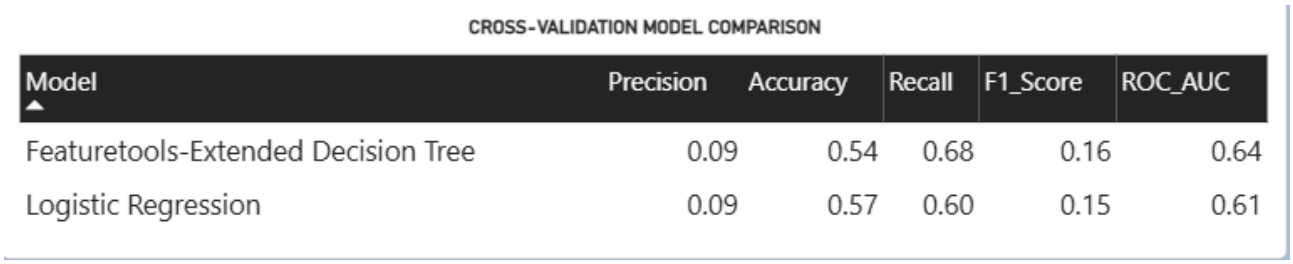
5.2 Logistic Regression assumptions and coefficient interpretation

Stage 2 checks whether policy IDs are unique and produces binned log-odds plots for selected continuous predictors. Unique policy IDs support a one-record-per-policy structure but do not prove statistical independence. Binned plots are a visual diagnostic and can conceal non-linearities.

The notebook's coefficient analysis identifies policy tenure and selected area clusters as associated with higher modelled claim odds, while age of car, population density and selected area clusters are associated with lower modelled odds. Because numerical predictors were standardised, numeric coefficients describe a one-standard-deviation change; categorical coefficients compare a category with its reference level. These are model associations, not causal effects.

5.3 Stage 3: Optuna-tuned Decision Tree

Stage 3 uses Featuretools-generated interactions and Optuna to tune a Decision Tree. The best saved trial reports 3-fold CV ROC-AUC of 0.6391 and parameters max_depth=3, min_samples_split=44 and min_samples_leaf=19. The final model is compared with Logistic Regression using five-fold stratified cross-validation.



CROSS-VALIDATION MODEL COMPARISON

Model	Precision	Accuracy	Recall	F1_Score	ROC_AUC
Featuretools-Extended Decision Tree	0.09	0.54	0.68	0.16	0.64
Logistic Regression	0.09	0.57	0.60	0.15	0.61

Figure 8. Cross-validation model comparison. The Featuretools-Extended Decision Tree has higher recall, F1 and ROC-AUC than Logistic Regression in the supplied comparison, while Logistic Regression has higher accuracy.

Model	Accuracy	Precision	Recall	F1 score	CV ROC-AUC
Featuretools-Extended Decision Tree	0.5412	0.0909	0.6835	0.1603	0.6365
Logistic Regression	0.5663	0.0859	0.5997	0.1503	0.6101

The Decision Tree is selected because its mean cross-validation ROC-AUC is higher (0.6365 vs 0.6101) and its recall and F1 are also higher. Logistic Regression retains higher accuracy. This trade-off is relevant in an imbalanced dataset: accuracy alone would favour the weaker claim-ranking model.

5.4 Held-out test evaluation

Selected model test metric	Result	Business meaning
ROC-AUC	0.6391	Some ranking ability above random; discrimination remains modest.
Recall	73.07%	About 73% of actual claim cases are identified at the selected classification threshold.
Precision	8.79%	Fewer than 1 in 10 flagged policies is an actual claim in the test set.
F1 score	0.157	Precision/recall balance remains low.
Accuracy	49.78%	Overall correctness is low and should not be used alone to judge this model.

The classification report shows 750 actual claim cases in the held-out test set and 10,969 no-claim cases. The selected model has claim-class recall of 0.73 and precision of 0.09 when rounded. High recall is potentially useful for triage, but low precision means substantial false-positive workload.

6. Predictive risk segmentation and business interpretation

6.1 Risk bands

The notebook creates predicted claim probabilities for the 11,719 held-out test policies and assigns Low, Medium, High and Very High bands. Because a shallow Decision Tree produces repeated probability scores, the notebook uses percentile ranks to handle ties. This means risk bands are relative to this scored test sample; they are not universal probability cutoffs and should not be presented as calibrated risk tiers without further validation.

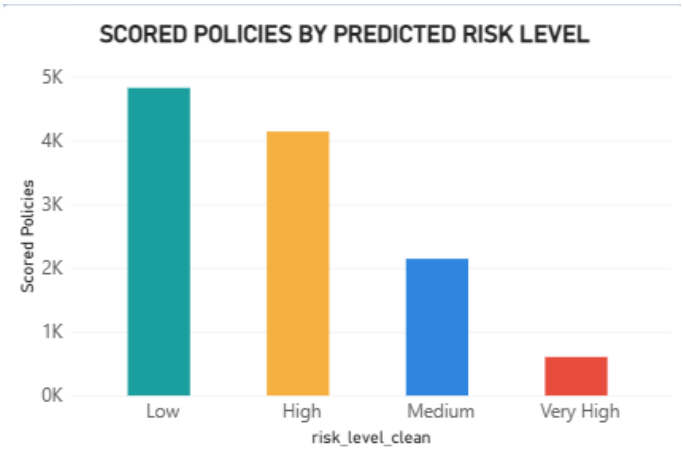


Figure 9. Scored policies by predicted risk level. The dashboard screenshot is a visual snapshot; the final notebook risk summary is the source of truth for the counts shown in the table below.

Risk band	Scored policies	Actual claims	Observed claim rate	Mean predicted probability
Low	4,826	167	3.46%	31.74%
Medium	2,145	134	6.25%	50.12%
High	4,142	362	8.74%	59.68%
Very High	606	87	14.36%	68.59%
Total	11,719	750	6.40%	—

6.2 Observed claim rate across risk bands

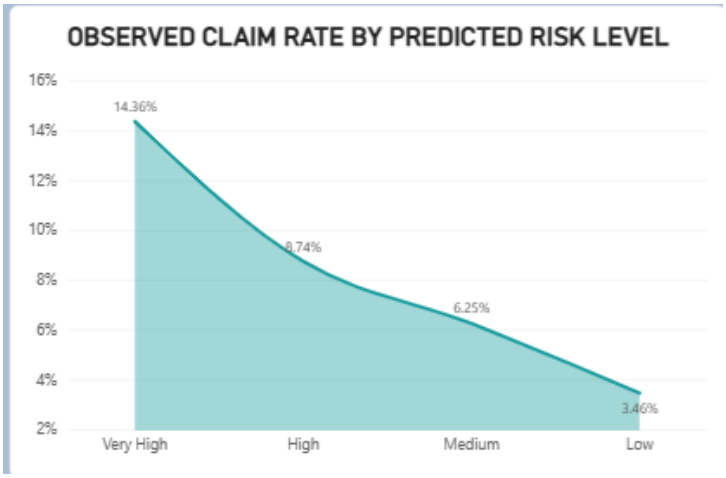


Figure 10. Observed claim rate by predicted risk level. The saved notebook reports a monotonic increase from 3.46% in Low to 14.36% in Very High, supporting relative prioritisation in this held-out sample.

The observed claim rate is about 2.24 times the portfolio rate in the Very High band (14.36% divided by 6.40%). This is a descriptive test-set lift, not a guarantee of future performance. Risk bands are derived from model probabilities and should be checked on future, unseen data before operational adoption.

6.3 Required probability distribution visual

The Stage 3 notebook also generates a boxplot of claim_probability by risk_level. The boxplot is intended to show the distribution, median and spread of predicted probabilities within each risk band. Include the notebook-generated boxplot in the final report/dashboard appendix and refresh it after the final model run. It complements the claim-rate chart: one visual shows model scores, while the other shows observed outcomes.

The supplied dashboard screenshot currently shows inconsistent category order/labels in one version of the risk-count visual. Before submission, ensure both risk visuals use the final exported risk_level field and show each category exactly once in the order Low, Medium, High, Very High.

7. Explainability, experiment tracking and governance

7.1 SHAP feature importance

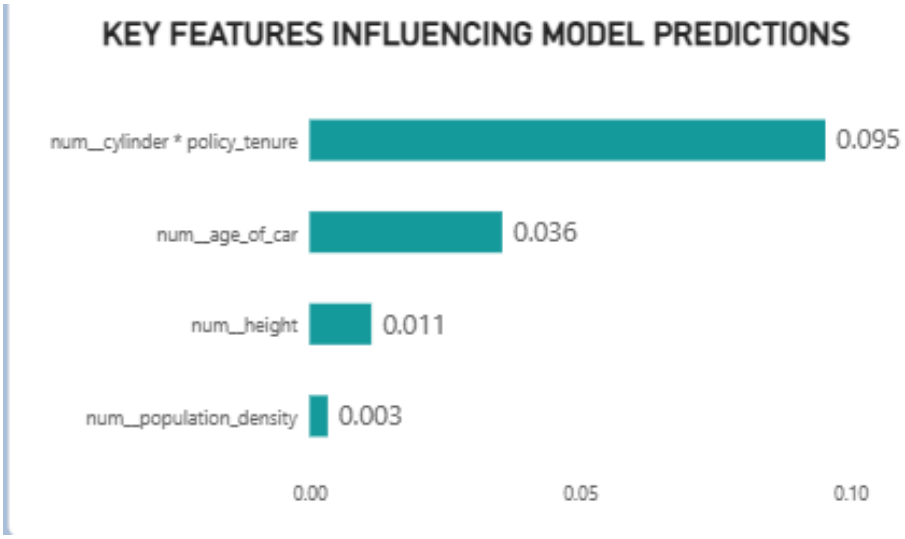


Figure 11. Key features influencing model predictions. The top feature is the Featuretools interaction cylinder × policy_tenure (mean absolute SHAP about 0.095), followed by age_of_car (0.036), height (0.011) and population_density (0.003).

The saved SHAP summary indicates that cylinder × policy_tenure contributes most strongly to predictions on average. The next features have much smaller mean absolute SHAP values. Several remaining features are reported as zero in the saved summary, so they should not be presented as meaningful drivers based on this model run.

SHAP values explain how the model allocates contributions to its outputs for the selected sample. They do not establish causal relationships, prove that a feature is a valid underwriting criterion, or guarantee the same ranking in future data.

7.2 Experiment tracking and model versioning

The notebook logs model experiments to MLflow and saves the selected pipeline as a versioned joblib artefact with metadata. The metadata is designed to record the model version, run identifier, training dataset, selected features, added Featuretools features, evaluation metrics and approval status. These files support traceability, but the saved candidate remains a research-stage model.

7.3 Data quality and drift report

The Stage 3 notebook uses Evidently to compare training and test predictor distributions and saves an HTML report. This is a train/test distribution check; it does not prove future stability, validate outcome performance over time, or replace production monitoring. The report should be retained with the notebook outputs if the rubric requires supporting artefacts.

7.4 Model approval status

The model should remain a candidate for further validation. The observed precision of 8.79% and ROC-AUC of 0.6391 are not sufficient evidence for automated underwriting decisions. Any future pilot should define review capacity, error costs, fairness controls, calibration requirements, and an agreed monitoring process before deployment.

8. Business recommendations and proposed operating workflow

8.1 Recommended use case

Use the score as a **human-review prioritisation aid**. A risk score can help order policies for investigation, but it should not automatically alter premiums, reject claims, or make eligibility decisions.

Step	Proposed action	Control / measure
1. Score	Generate probabilities for eligible policies using a versioned model.	Record model version, scoring date and input data version.
2. Prioritise	Rank policies within the review population; agree a review-capacity threshold with underwriting.	Measure the number of policies flagged and share of observed claims captured.
3. Review	A trained reviewer checks supporting evidence and applies existing policy rules.	Keep a human decision-maker; record reasons and overrides.
4. Evaluate	Compare outcomes for reviewed and non-reviewed groups where the evaluation design allows.	Track precision, recall, lift, false-positive workload and missed claims.
5. Monitor	Review input drift and outcome performance on a scheduled cadence.	Use drift alerts as prompts for investigation, not proof of model failure.
6. Revalidate	Reassess calibration, stability and fairness before changing model or policy rules.	Require formal approval and documented validation.

8.2 Priority recommendations

- Set an operational threshold based on underwriting review capacity and the relative cost of missed claims versus false positives.
- Evaluate precision and claim capture in the top 5%, 10%, 20% and other review-capacity bands rather than relying only on the default 0.5 classification threshold.
- Assess probability calibration before treating claim_probability as an absolute likelihood. Current risk bands are rank-based because tree probability values are tied.
- Validate performance prospectively and across policy cohorts, vehicle segments, NCAP categories and relevant area clusters.
- Review fairness, stability and appropriateness of features with governance stakeholders before any operational use.
- Monitor data quality and drift, and document retraining triggers, rollback procedures and ownership.

8.3 What the analysis does not prove

- It does not prove that a specific vehicle characteristic causes a claim.
- It does not quantify the value of avoided payouts or savings because financial cost data and an impact evaluation were not supplied.
- It does not show actual speeding incidents; the available field is speed-alert system availability.
- It does not establish production readiness or generalisation beyond the supplied train/test split.

9. Limitations, quality controls and conclusion

9.1 Limitations

- Class imbalance: claims are 6.40% of the data, so performance depends on the error trade-off and the selected threshold.
- Low precision: 8.79% precision means the majority of policies flagged as claims are false positives at the evaluated threshold.
- Moderate ranking performance: test ROC-AUC is 0.6391, indicating improvement over random ranking but limited discrimination.
- Risk bands are relative percentile ranks for the held-out sample and are not calibrated business risk definitions.
- Small subgroup: the Older group in the supplied three-level vehicle-age visual has only 14 policies; the displayed 0% claim rate is unstable.
- Speed-alert interpretation: availability is not the same as speeding behaviour; group sizes are imbalanced.
- Observational data: findings may reflect confounding and should not be treated as causal.
- Data/export alignment: ensure the Power BI CSVs are regenerated from the final notebook run so the visual counts and rates match the saved risk summary.

9.2 Final quality-control checklist

Check	Required final action
Notebook reproducibility	Run Stage 1, Stage 2 and Stage 3 sequentially from a clean kernel and save outputs.
Model comparison	Use CV metrics for model selection and report held-out test metrics separately.
Risk band data	Refresh prediction_output.csv and all Power BI exports from the final Stage 3 run.
Dashboard labels	Both risk charts must show Low, Medium, High and Very High exactly once and in the same order.
Model comparison table	Keep metric columns unaggregated; remove totals; correct the title to CROSS-VALIDATION MODEL COMPARISON.
Probability boxplot	Include claim_probability by risk_level in the report appendix or dashboard appendix.
Export inspection	Open dashboard_report.pdf and inspect every page for clipping, overlap, and unreadable values.
Supporting artefacts	Retain MLflow log, model metadata, Evidently report and SHAP summary if required by submission instructions.

9.3 Conclusion

The SafeDrive capstone establishes an end-to-end workflow from data inspection and feature engineering through baseline modelling, AutoML-style feature generation and tuning, evaluation, explainability and risk segmentation. The Featuretools-Extended Decision Tree outperforms Logistic Regression on cross-validation ROC-AUC (0.6365 vs 0.6101) and achieves 0.6391 ROC-AUC on the held-out test set. The risk bands in the saved notebook show increasing observed claim rates from 3.46% in Low to 14.36% in Very High.

The results support further investigation of a controlled human-review prioritisation use case, but not automated underwriting. Low precision, modest ROC-AUC, the class imbalance and the need to reconcile Power BI exports are material constraints. The appropriate next step is a documented, prospectively monitored pilot only after the final artefacts are synchronised and governance criteria are met.

This report is based on the supplied Stage 1, Stage 2 and Stage 3 notebooks and the dashboard images provided in the conversation. It does not independently verify unprovided business data, financial impact, or future performance.

Appendix A — Dashboard visual gallery

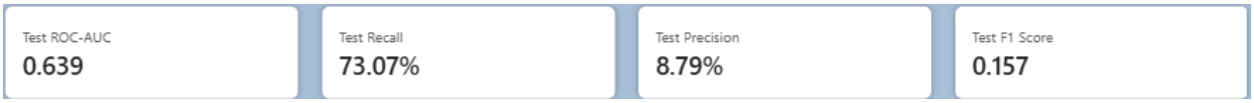
The following gallery preserves every dashboard screenshot supplied for this report and gives a concise description of what each visual communicates. These are supplied visual captures, not reconstructed charts. Review them against the refreshed Power BI export before final submission.

A.1 Portfolio KPI cards



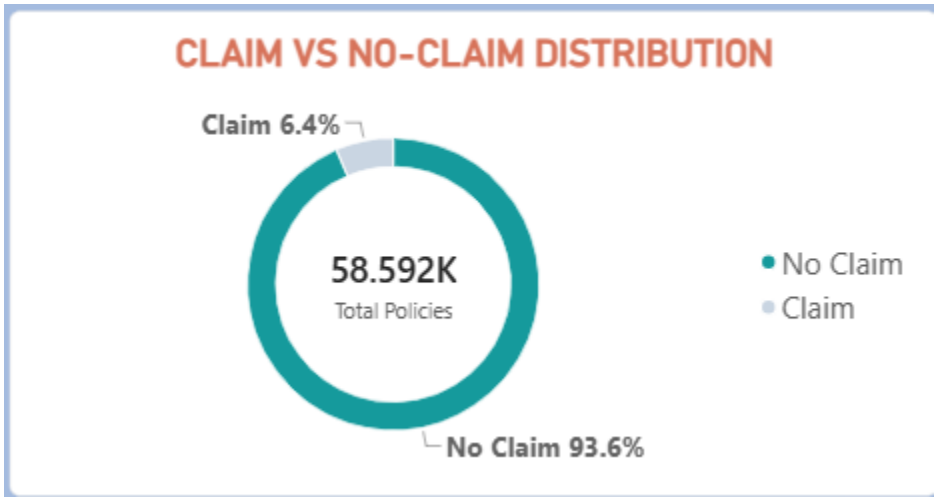
Summarises portfolio size, claims, overall claim rate, scored policy count and average NCAP rating.

A.2 Model performance KPI cards



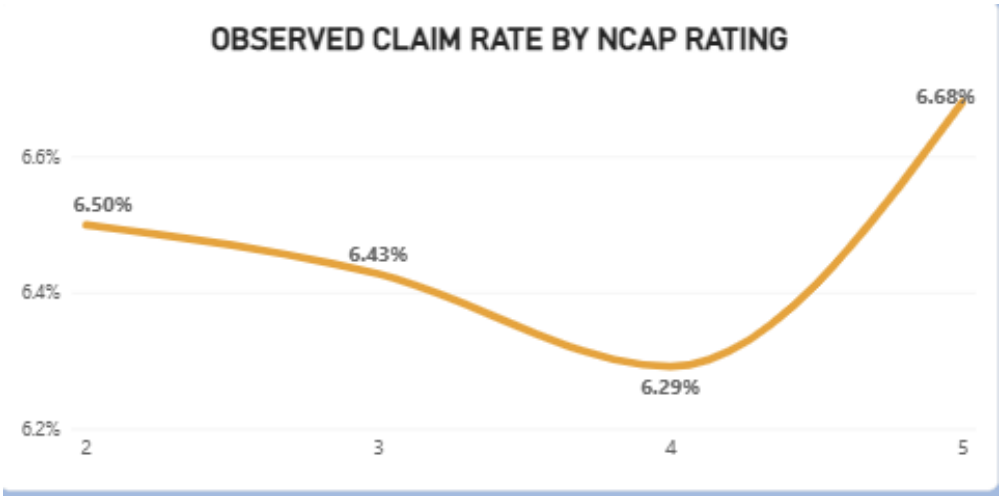
Summarises the selected model's test ROC-AUC, recall, precision and F1 score.

A.3 Claim vs no-claim distribution



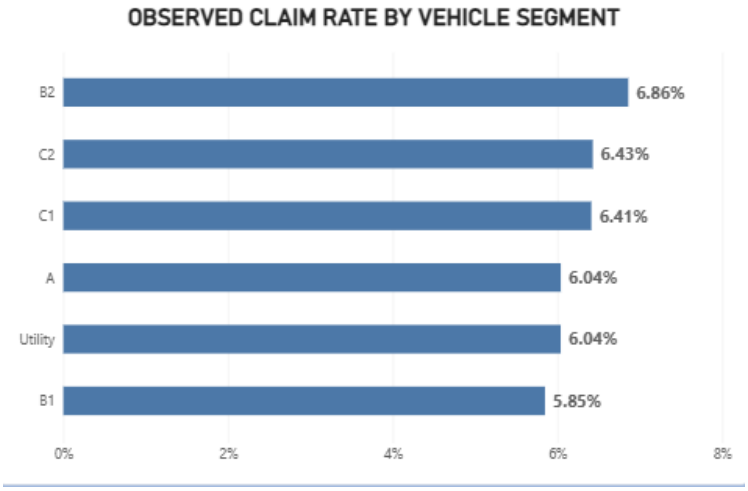
Shows the portfolio class imbalance: 6.4% claims and 93.6% no claims.

A.4 Observed claim rate by NCAP rating



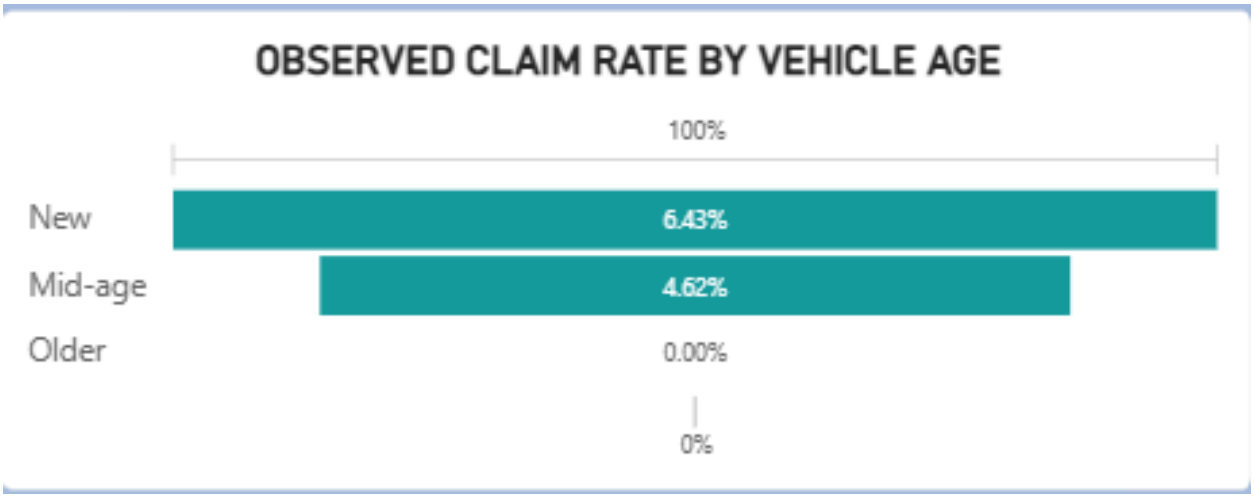
Compares observed claim rates across NCAP ratings 2 to 5.

A.5 Observed claim rate by vehicle segment



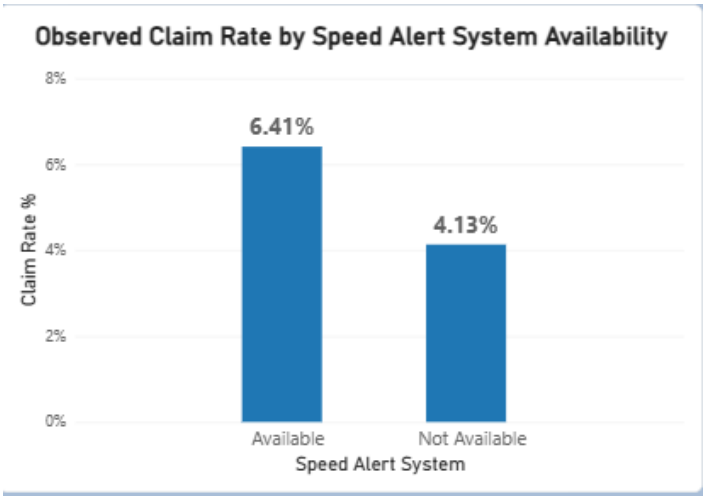
Ranks the displayed vehicle segments by observed claim rate.

A.6 Observed claim rate by vehicle age



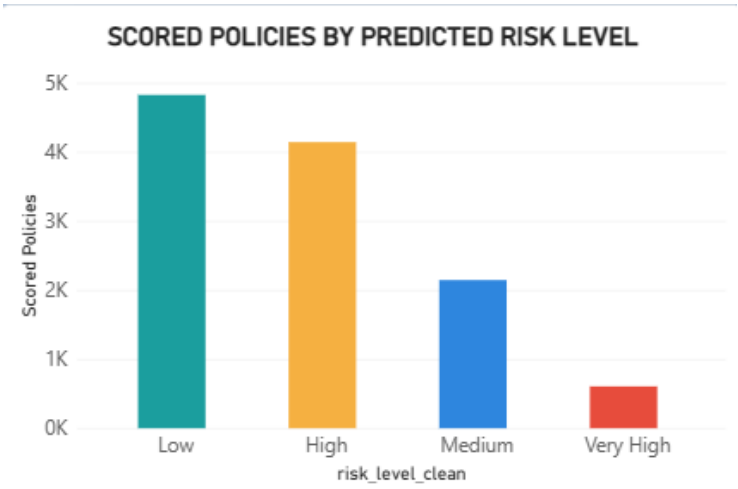
Compares New, Mid-age and Older group rates; Older group has a very small sample.

A.7 Claim rate by speed-alert availability



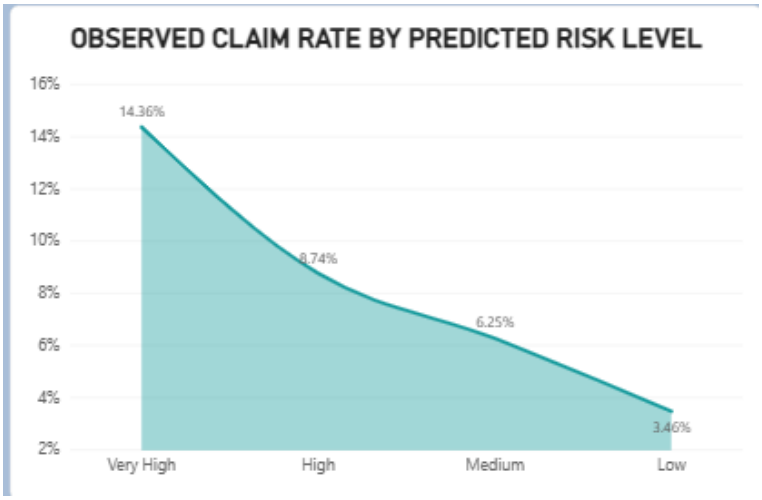
Compares observed claim rates where the system is available versus not available.

A.8 Scored policies by risk level



Shows the count of held-out policies in each predicted risk band; category order must match the final notebook output.

A.9 Observed claim rate by predicted risk level



Shows observed claim rates increasing across the four notebook risk bands.

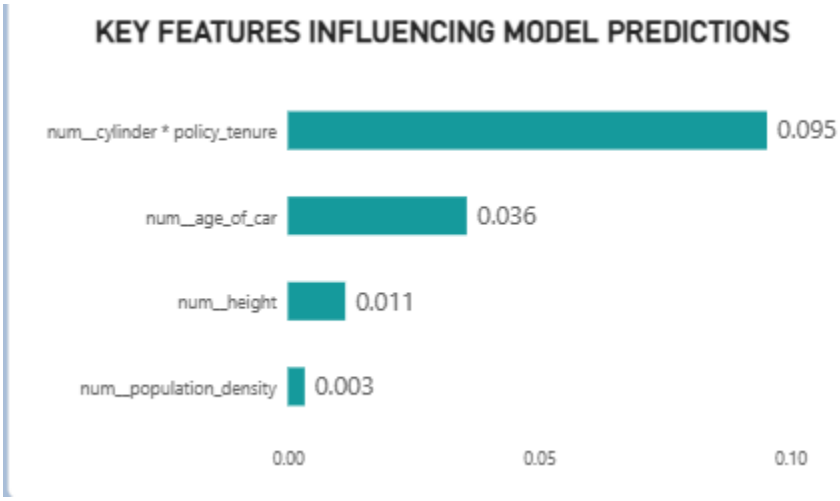
A.10 Cross-validation model comparison

CROSS-VALIDATION MODEL COMPARISON

Model	Precision	Accuracy	Recall	F1_Score	ROC_AUC
Featuretools-Extended Decision Tree	0.09	0.54	0.68	0.16	0.64
Logistic Regression	0.09	0.57	0.60	0.15	0.61

Compares the Decision Tree and Logistic Regression using Accuracy, Precision, Recall, F1 and ROC-AUC.

A.11 Key features influencing model predictions



Displays the leading mean absolute SHAP values, led by cylinder × policy_tenure.